IMMUNICARE: A Decision Support System for Infant Immunization Tracking and Monitoring with Spatial Clustering using DBSCAN

A Research Project Presented to the San Pedro City Health Office

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Table of Contents

Chapter 1

THE PROBLEM AND ITS SETTING

The global landscape of public health has seen a significant trend toward the digitalization of immunization records to combat the persistent threat of vaccine-preventable diseases. Across various developing nations, there is an increasing shift from traditional paper-based documentation to electronic immunization registries. These modern systems are designed to enhance data accuracy and provide real-time insights into community health status (Ahasan et al., 2022). Recent advancements in Geographic Information Systems (GIS) and spatial data science have further transformed how health authorities visualize population coverage, allowing for a more granular understanding of where health services are succeeding or failing at a local level (Fallucca et al., 2024).

However, a critical issue persists within the context of Rural Health Units (RHUs) in the Philippines. Despite the availability of digital tools, many local health facilities still rely on fragmented manual tracking procedures that are prone to human error and data loss. This lack of a centralized and automated tracking mechanism leads to significant delays in identifying infants who have missed their scheduled doses (Rahmadhan et al., 2023). Without a streamlined digital record, health workers struggle to maintain an organized follow-up system, resulting in a growing number of defaulters who remain unprotected against life-threatening illnesses.

This gap presents a significant opportunity for the integration of advanced spatial analytics into local health informatics. By moving beyond simple digital logs and adopting spatial clustering algorithms, such as the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm, there is a chance to transform raw geographic data into actionable intelligence. This technology allows for the identification of specific pockets of vulnerability — specifically, areas where unvaccinated infants are concentrated — which are often invisible in standard statistical reports (PLOS Global Public Health, 2024). Leveraging these analytical tools can bridge the divide between data collection and field implementation, providing a more sophisticated approach to public health management.

The primary challenge remains the operational environment of rural healthcare. Barangay Health Workers (BHWs) are often burdened with heavy workloads and limited transportation resources, making random house-to-house visits inefficient and physically exhausting. Furthermore, many existing immunization systems are designed for national-level monitoring and are not optimized for the specific needs of rural personnel who require precise coordinates to plan their daily activities (Yang et al., 2025). There is a pressing need for a system that is not only technologically advanced but also functionally suitable and reliable under the constraints of limited manpower and rural infrastructure.

Despite the growing body of literature on digital immunization systems and spatial analytics in public health, a critical research gap exists at the intersection of these two domains when applied to rural health unit settings in the Philippines. Existing immunization information systems are predominantly designed for national-level monitoring and lack the localized analytical capability required for community-level decision support. While geospatial heatmaps provide descriptive visualization of coverage disparities, they do not mathematically distinguish between isolated cases and high-density clusters of unvaccinated infants — a distinction that is essential for efficient outreach planning under conditions of limited manpower and transportation. Furthermore, quantitative evidence on the integration of density-based spatial clustering algorithms, specifically DBSCAN, within a web-based decision support system designed for frontline health workers in rural Philippine settings is absent from the literature. This study directly addresses that gap by developing and evaluating IMMUNICARE — a system that moves beyond descriptive mapping and into prescriptive, cluster-based decision support — within the operational and infrastructural constraints of a single RHU in San Pedro City, Laguna.

Therefore, the purpose of this study is to develop and evaluate IMMUNICARE, a spatial cluster-based Decision Support System (DSS) designed specifically for Rural Health Units. Grounded in the foundational framework of Sprague and Carlson (1982), which defines a DSS as a computer-based tool that supports semi-structured decision-making through integrated data management, model management, and user interface subsystems, IMMUNICARE operationalizes these components through its vaccination records database, DBSCAN-driven scheduling engine, and geospatial dashboard, respectively. The system aims to automate the tracking of infant vaccination schedules and utilize spatial clustering to identify high-priority action zones. By providing precise geographical centroids for outreach, the research intends to optimize the deployment of health resources and ensure that immunization efforts are targeted toward the most underserved populations in the community.

Statement of the Problem

Despite the availability of routine immunization services in rural health units, the timely and complete vaccination of infants within their first year of life remains a challenge due to the continued use of manual, paper-based record-keeping systems. These systems result in incomplete records, difficulty identifying infants who are due or overdue for vaccination, a high number of defaulters, and the absence of systematic reminder mechanisms for caregivers. Additionally, existing practices often lack secure access control mechanisms, exposing sensitive patient information to potential unauthorized access. While web-based immunization monitoring systems offer a potential solution, limited quantitative evidence exists on their effectiveness in improving immunization data management, generating actionable insights, and supporting data-driven decision-making for vaccination outreach in rural health unit settings.

The following questions shall be answered in this research paper:

1. How do automated SMS reminders serve as cues to action in improving caregiver vaccination attendance and compliance based on the Health Belief Model (HBM)?
2. What is the level of technological acceptance of IMMUNICARE among RHU staff based on the UTAUT model?
   - 2.1. Performance expectancy;
   - 2.2. Effort expectancy;
   - 2.3. Social influence;
   - 2.4. Facilitating conditions;

2.5. Behavioral Intention to Use?

3. What is the level of effectiveness of IMMUNICARE based on the ISO/IEC 25010 standard in terms of:

   3.1. Functional Suitability;

   3.2. Usability;

   3.3. Performance Efficiency;

   3.4. Reliability; and

   3.5. Security?

THEORETICAL FRAMEWORK

This study is grounded on complementary behavioral and technology adoption theories that explain how IMMUNICARE, a Web Based Infant Immunization Tracking and Decision Support System with Spatial Clustering for Vaccination Outreach, functions at the caregiver and health worker levels. These theories provide the explanatory basis for understanding vaccination adherence among caregivers and system adoption among frontline health personnel in Rural Health Units. Together, they guide the design of key system features including automated scheduling, SMS reminders, and DBSCAN based spatial clustering, and inform the selection of variables used to quantitatively evaluate system effectiveness.

Health Belief Model

The Health Belief Model (HBM) explains caregiver adherence to infant vaccination schedules by examining factors that influence preventive health behavior, including perceived susceptibility, perceived severity, perceived benefits, perceived barriers, cues to action, and self-efficacy. Evidence from a study involving 220 parents in Saudi Arabia demonstrated that non-adherence to childhood vaccination schedules was strongly associated with higher perceived barriers, lower perceived disease risk, and reduced confidence in managing vaccination schedules, with forgetting appointments identified as the most common practical challenge (Hobani & Alhalal, 2022).

These findings reflect conditions commonly observed in rural health units, where caregivers often intend to return for follow-up immunizations but miss appointments due to forgetfulness or lack of reminders. In this study, the Health Belief Model informs the design of automated SMS reminders and informational messages within the system. These features serve as cues to action, reduce practical barriers through clear scheduling information, and support caregiver self-efficacy. Quantitatively, the model underpins the measurement of caregiver compliance and vaccination attendance rates before and after system implementation.

Unified Theory of Acceptance and Use of Technology (UTAUT)

The Unified Theory of Acceptance and Use of Technology (UTAUT) is used to explain the adoption and sustained use of the system by health care professionals. UTAUT proposes that technology use is influenced by performance expectancy, effort expectancy, social influence, and facilitating conditions, which together shape behavioral intention and actual system usage (Miguel Cruz et al., 2022).

Empirical evidence from a pilot study involving health care aides found that performance expectancy was the strongest predictor of intention to use a digital health application, with behavioral intention significantly predicting continued use over time (Miguel Cruz et al., 2022). Guided by this framework, IMMUNICARE is designed to enable midwives and barangay health workers to quickly identify infants who are due or overdue for vaccination while integrating seamlessly into existing immunization workflows. The framework also informs quantitative evaluation measures related to perceived usefulness, ease of use, and intention to continue using the system.

Spatial Epidemiology

Spatial Epidemiology provides the theoretical foundation for the geospatial heatmap visualization feature of IMMUNICARE. Spatial epidemiology is defined as the study of spatial variation in disease risk or health outcomes, aimed at identifying geographic risk factors and populations requiring targeted public health intervention (Elliott & Wartenberg, 2004). Within this framework, unvaccinated and under-vaccinated infants are not assumed to be randomly distributed across a community but are understood to concentrate in specific geographic areas referred to as hotspots that can be identified, visualized, and acted upon using geospatial analysis methods (Cromley & McLafferty, 2024).

In the context of IMMUNICARE, Spatial Epidemiology directly informs the design of the geospatial heatmap visualization feature of the system, which translates vaccination record data into a color coded spatial display. This allows health workers and barangay health workers at RHU in San Pedro City to visually identify purok level concentrations of unreached infants, enabling them to prioritize outreach activities in areas with the highest immunization gaps. The framework supports the premise that making geographic immunization inequities visible to frontline health personnel is a critical step toward more efficient, evidence based, and targeted vaccination outreach (Siddique et al., 2023).

Systems and Software Quality (ISO/IEC 25010)

The ISO/IEC 25010 standard provides the technical framework for evaluating software product quality through a structured hierarchy of characteristics including functional suitability, performance efficiency, usability, reliability, and security (ISO/IEC, 2011). While behavioral models like UTAUT focus on the human user, the ISO/IEC 25010 standard ensures that the technical architecture of the system meets objective performance benchmarks. Research utilizing this standard to evaluate integrated health information systems found that high scores in functional suitability and reliability are non-negotiable requirements for systems handling critical patient data, as technical failures in these areas can lead to clinical errors and loss of user trust (Mulyani et al., 2022).

In this study, ISO/IEC 25010 justifies the quantitative technical assessment of IMMUNICARE's backend and frontend performance. It provides the criteria used by IT experts to verify that the system's automated scheduling logic is functionally correct and that the database architecture is secure and reliable under operational stress. By integrating this standard, the study ensures that the system is not only accepted by users but is also technically robust enough to function as a dependable tool for public health governance in San Pedro City.

CONCEPTUAL FRAMEWORK

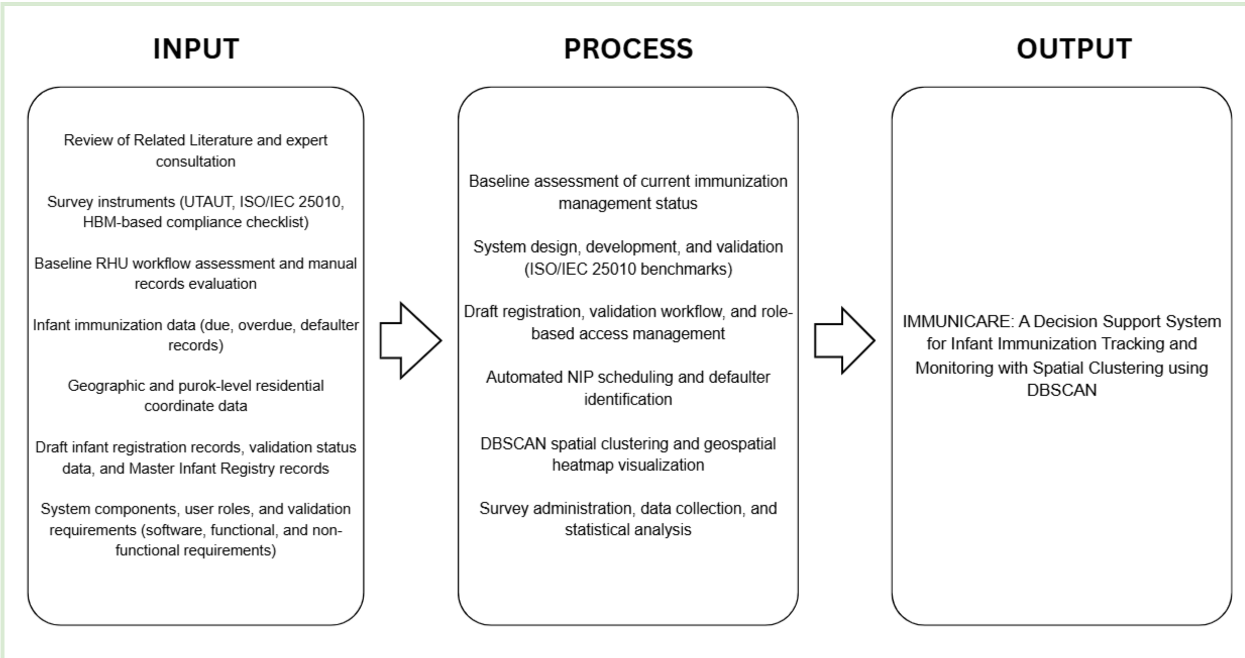


Figure 1: Conceptual Framework

The conceptual framework of this study follows the Input–Process–Output (IPO) model and integrates five theoretical anchors — the Decision Support System (DSS) framework, the Health Belief Model (HBM), the Unified Theory of Acceptance and Use of Technology (UTAUT), Spatial Epidemiology, and ISO/IEC 25010 — across its three layers. The framework illustrates how research inputs, operational immunization data, and system requirements are transformed through a set of structured, theory-driven processes into the proposed output of the study: IMMUNICARE, a Decision Support System for Infant Immunization Tracking and Monitoring with Spatial Clustering using DBSCAN.

The Input layer consists of both research-related and system-related components. The research inputs include the review of related literature on immunization systems, spatial clustering, SMS reminder interventions, community health worker workflows, and health informatics decision support systems; survey instruments based on UTAUT, ISO/IEC 25010, and HBM; baseline assessment of current immunization workflows and manual records at the Rural Health Unit (RHU); and expert consultation for content validation of the system and evaluation tools. The system-related inputs include infant immunization records such as due, overdue, and defaulter cases; geographic and purok-level coordinate data; caregiver information for follow-up and SMS notification purposes; Draft infant registration records, validation status records, and Master Infant Registry data; and system requirements involving software components, user roles, validation workflows, and role-based access structures.

The Process layer represents the transformation of these inputs into functional system operations and measurable research outcomes. The DSS framework governs the integration of data management, schedule computation, clustering analysis, and decision-support workflows within the proposed system. The Health Belief Model guides the automated SMS reminder process by treating reminders as cues to action that encourage caregiver compliance with vaccination schedules. UTAUT governs the design and evaluation of system usability and user acceptance among RHU staff, particularly in relation to performance expectancy, effort expectancy, and behavioral intention to use the system. Spatial Epidemiology governs the DBSCAN clustering process by identifying non-random geographic concentrations of under-immunized infants at the barangay and purok levels. ISO/IEC 25010 governs the design validation and quality evaluation of the system across the dimensions of functional suitability, usability, performance efficiency, reliability, and security.

Operationally, the Process layer also includes baseline assessment of current immunization management practices, system design and development, Draft registration and validation workflows, role-based access management, automated National Immunization Program (NIP) schedule generation, defaulter identification, DBSCAN spatial clustering, geospatial heatmap visualization, survey administration, data collection, and statistical analysis. Within the revised workflow, Barangay Health Workers (BHWs) create Draft infant registration records which undergo Midwife validation before promotion into the official Master Infant Registry. Only validated records are processed for schedule generation, clustering analysis, reporting, and monitoring functions. The system likewise implements barangay-scoped role-based access control, audit logging, and OTP-authenticated caregiver portal access to support data integrity, accountability, and information security.

The Output layer is represented by the completed IMMUNICARE system itself. All inputs processed through the defined system and research processes converge to produce a functional, evaluated, and theoretically grounded Decision Support System for infant immunization tracking and monitoring. The completed system, together with its evaluation results under the UTAUT, HBM, and ISO/IEC 25010 frameworks, constitutes the totality of the study's output.

SCOPE AND DELIMITATION OF THE STUDY

This study encompasses both the development and the quantitative evaluation of IMMUNICARE, a Decision Support System for infant immunization tracking and monitoring with spatial clustering using DBSCAN. IMMUNICARE is designed to support infant immunization monitoring and decision support workflows across Rural Health Units within San Pedro City, Laguna. However, due to time and operational constraints, the pilot implementation and evaluation of the proposed system are limited to selected Rural Health Unit sites during the conduct of the study. The study covers infants from birth to twelve (12) months of age, consistent with the complete first-year National Immunization Program (NIP) schedule.

The respondents for the UTAUT-based acceptance evaluation are limited to RHU staff directly involved in immunization workflows, specifically Super Admin / Head Nurses, Midwives, Barangay Health Workers (BHWs), and authorized system users assigned at the pilot site. The respondents for the ISO/IEC 25010 quality evaluation are IT experts with relevant experience in software testing or health information systems. Caregiver respondents for the HBM-based survey instrument are purposely selected from among registered caregivers of infants enrolled in the National Immunization Program at the pilot RHU during the post-implementation period.

Caregiver data for Research Question 2 are collected through two complementary sources: objective vaccination attendance records maintained at the pilot RHU and a structured Likert-scale self-report questionnaire administered directly to caregivers. This dual-source approach is intentional and methodologically grounded. Attendance records capture actual compliance behavior—whether caregivers brought their child for vaccination on schedule—while the caregiver survey captures the perceptual and attitudinal factors associated with that behavior, including cues to action, perceived benefits, and self-reported adherence. Behavior alone is insufficient to explain health decisions; understanding the cognitive processes that mediate compliance requires direct input from caregivers themselves. The integration of both data types strengthens the study's explanatory power and aligns fully with the Health Belief Model framework.

The study does not assess long-term immunization outcomes beyond the evaluation period, nor does it conduct a cost-effectiveness analysis. Findings are not generalized beyond the pilot RHU site, and the study does not compare IMMUNICARE against other existing digital immunization systems in a controlled experimental design. The system does not include modules for vaccine inventory management, stock forecasting, or cold chain monitoring, as these fall outside the scope of the current evaluation. IMMUNICARE has not been integrated with national-level immunization databases or other external health information systems.

In terms of system scope, IMMUNICARE supports the operational workflows of Super Admin / Head Nurses, Midwives, Barangay Health Workers (BHWs), and caregivers through a barangay-scoped role-based access structure. The system components are organized into functional modules that support infant registration, immunization monitoring, validation workflows, clustering analysis, notification management, reporting, and audit tracking.

In terms of software requirements, IMMUNICARE is a web-based platform accessible through standard browsers including Chrome, Firefox, and Edge. The system requires a stable internet connection for real-time data entry, SMS notification dispatch, synchronization, and spatial clustering computation. The proposed system does not include offline functionality or native mobile application support during the scope of this study. The system is hosted on a server with a database management system that handles immunization records, caregiver information, and geographic data.

In terms of functional requirements, the system includes: (1) an infant registration and profiling module that records infant demographics, caregiver details, maternal vaccination status, birth dose records, and residential coordinates; (2) a Draft registration and validation workflow in which BHW-submitted records undergo Midwife review before promotion to the official Master Infant Registry; (3) an automated NIP schedule engine that computes vaccine due dates based on the infant's date of birth, enforces minimum dose interval rules, and applies catch-up logic when doses are missed; (4) a vaccination recording and monitoring module that allows health workers to identify infants who are due or overdue; (5) a DBSCAN spatial clustering module with a geospatial heatmap dashboard that groups under-immunized infants into geographic clusters and displays results at the barangay and purok levels; (6) an automated SMS notification module that sends vaccination reminders and alerts to caregivers; (7) a Department of Health report generation module for Monthly Form 1 (M1) and Annual Form A1; and (8) role-based access control for Super Admin / Head Nurses, Midwives, Barangay Health Workers, and caregivers.

The system implements a barangay-scoped role-based access structure in which users may only access records within their assigned area of responsibility. Barangay Health Workers are responsible for creating Draft infant registration records and submitting them for Midwife validation. The Midwife reviews submitted records and may approve, return for correction, or reject the registration before it becomes part of the official Master Infant Registry. Only approved records are eligible for immunization schedule generation, vaccination recording, clustering analysis, and reporting functions. The system also supports audit logging, caregiver OTP-authenticated portal access, and role-specific dashboards for monitoring immunization activities.

In terms of non-functional requirements, the system is designed to meet the quality benchmarks of ISO/IEC 25010 across the dimensions of functional suitability, usability, performance efficiency, reliability, and security. Security is implemented through role-based access control that defines distinct permission levels for all authorized user types. Geospatial visualizations and clustering results are restricted to aggregated barangay and purok levels to maintain individual data privacy, with no household-level GPS coordinates exposed to unauthorized users. The system is designed for performance that supports schedule generation, validation processing, and cluster computation within acceptable operational load times during RHU working hours.

Several factors further define the delimitation of this study. The accuracy of vaccination records, clustering results, and generated reports depends entirely on correct data entry and coordinate validation by authorized health personnel. The system is intended only for infant immunization monitoring and does not cover adolescent or adult vaccination programs. These boundaries define the context of the study and guide the interpretation of all quantitative results.

SIGNIFICANCE OF THE STUDY

This study is expected to benefit the following stakeholders:

Infants and Families. Caregivers often experience difficulty remembering scheduled vaccination dates, particularly during the first year of life when multiple doses are required and when infants appear healthy. The implementation of a web-based immunization monitoring system with automated SMS reminders provides caregivers with timely and accurate information regarding upcoming and missed vaccinations. By supporting reminder-based adherence and enabling health workers to identify infants who have missed scheduled doses, the system is expected to contribute to improved completion of infant immunization schedules and increased protection against vaccine-preventable diseases.

Midwives and Barangay Health Workers. Midwives and barangay health workers in rural health units commonly rely on manual logbooks to determine which infants are due or overdue for vaccination, a process that is time-consuming and prone to error. The proposed system consolidates immunization records into a single digital platform, allowing health workers to efficiently track vaccination status, identify defaulters, and prioritize follow-up activities. The inclusion of location-based visualization further supports outreach planning by highlighting areas with higher concentrations of delayed or missed vaccinations.

Rural Health Unit and Local Government. At the facility and local government levels, the system provides timely and consolidated immunization data that support evidence-based planning, monitoring, and reporting. Aggregated barangay-level information enables rural health units to assess immunization coverage more accurately, identify gaps in service delivery, and allocate outreach resources more effectively. As a low-cost, web-based solution designed for existing infrastructure, the system may also support improved compliance with Department of Health reporting requirements without imposing significant additional operational burden.

Academic and Research Community. Most existing studies on immunization information systems focus on national programs or hospital-based implementations, with limited empirical evidence from rural health unit settings. This study contributes quantitative evidence on the effectiveness of a web-based infant immunization monitoring system implemented at the community level. The findings may inform future research on digital health interventions designed for low-resource primary care environments and guide the evaluation of similar systems.

Future Health Information System Development. The design and evaluation of IMMUNICARE may serve as a reference model for other community-based digital health initiatives, such as maternal health monitoring or treatment adherence tracking. By demonstrating how a focused, workflow-aligned system can support frontline health workers and caregivers, the study highlights the potential of scalable, low-cost digital solutions to strengthen essential public health services in rural settings.

DEFINITION OF TERMS

IMMUNICARE. The web-based infant immunization tracking and decision support system developed in this study, designed to improve vaccination tracking, defaulter identification, caregiver compliance, and outreach planning in rural health units through automated scheduling, SMS reminders, geospatial heatmap visualization, and DBSCAN based spatial clustering for vaccination outreach.

Automated NIP Schedule Engine. A system feature of IMMUNICARE that automatically computes vaccine due dates based on an infant's date of birth following the National Immunization Program schedule, enforces minimum dose intervals, and applies catch-up logic when doses are missed.

DBSCAN (Density-Based Spatial Clustering of Applications with Noise). A spatial clustering algorithm used in IMMUNICARE to mathematically identify geographic groupings of under-immunized infants based on the density of their residential coordinates . DBSCAN does not require a predefined number of clusters, it is capable of detecting clusters of arbitrary shapes, and automatically classifies isolated cases as noise. This allows the system to filter out outliers and focus outreach resources on significant, high-density concentrations of overdue vaccinations .

DOH-Standard Immunization Forms. The official formats prescribed by the Department of Health that the system's reporting module follows when generating monthly (M1) and annual (A1) vaccination summaries for submission to higher health offices.

Dropouts / Defaulters in the Vaccination Program. Infants who begin their vaccination schedule but stop attending before completing all required doses, often due to forgotten appointments, lost records, or families moving without proper transfer of care.

Geographical Centroid. The mathematically calculated central point of a specific spatial cluster identified by the system . In this research, the centroid serves as a strategic "anchor point" for planning mobile vaccination clinics or community outreach, ensuring that health resources are deployed at the most accessible center-point of an underserved infant population .

Geospatial Heatmap. A map-based visualization technique used in IMMUNICARE that represents the geographic distribution and density of under-immunized or unvaccinated infants through a color-coded spatial display. Areas with higher concentrations of missed or delayed vaccinations are rendered in warmer colors, allowing health workers to visually identify immunization gaps at the barangay and purok levels without requiring advanced data analysis skills.

One-Time Password (OTP). A temporary, single-use authentication code sent to a registered mobile number, used in IMMUNICARE to provide caregivers with secure, limited access to their child's vaccination record without requiring the creation of a permanent account. This mechanism supports data privacy by ensuring that sensitive health information is accessible only to the verified caregiver associated with a specific infant record.

Role-Based Access Control (RBAC). A security mechanism implemented in IMMUNICARE that restricts system access and functionality based on the assigned role of each user. The system defines distinct permission levels for administrators, midwives or nurses, Barangay Health Workers, and caregivers, ensuring that sensitive immunization data is accessible only to authorized personnel appropriate to their function within the Rural Health Unit.

Spatial Clustering. A data mining technique used in this study to group infants based on the proximity of their residential coordinates . Unlike simple mapping, spatial clustering identifies non-random patterns of under-immunization, allowing health authorities to distinguish between isolated cases and high-density geographic areas that require immediate intervention.

Zero-Dose and Under-Immunized Children. Zero-dose children are infants who have not received any of the basic routine vaccinations, while under-immunized children are those who started but did not complete their vaccination schedule, leaving them vulnerable to vaccine-preventable diseases.

Chapter 2

REVIEW OF RELATED LITERATURE

This chapter evaluates the recent publications on the subject of routine infant immunization and global coverage disparities, determinants of missed and incomplete vaccination, the role of digital health and SMS reminder interventions, community health worker-centered digital support systems, geospatial technologies and equity in immunization, spatial data mining and clustering algorithms in public health decision support, and immunization information systems including data use and privacy considerations.

Global Trends and Disparities in Infant Immunization

Although global immunization coverage has improved over the years in immunization coverage, recently, the rate of global coverage has plateaued for millions of children who are still inadequately vaccinated. Currently, available statistics show that there are 14.3 million zero-dose children worldwide who are unvaccinated for their routine vaccinations, while in many countries in Asia and sub-Saharan Africa, the rate of dropout between doses of routine vaccinations has largely surpassed the accepted levels. In Ethiopia, for example, dropout between BCG/Measles vaccine doses of 16.9% has been recorded, while in some areas, dropout levels surpass 40%, which has been similar for Nigeria and other countries of similar context. (World Health Organization, 2024; Golla et al., 2025)

Determinants of Missed, Delayed, and Incomplete Vaccination

Studies investigating the reasons for caregivers missing or postponing vaccines indicate that forgetfulness and practical obstacles are the major factors, while vaccine hesitancy and refusal play only a minor role. The application of the Health Belief Model in studies suggests that more than fifty percent of parents face at least one instance of their child's vaccine being missed or delayed. The primary reason for this is the mere forgetting of the appointment dates; among the others, the concerns regarding the accessibility of services and the low self-efficacy in keeping the child's vaccination schedule are also significant. The areas of low vaccination rate also are influenced by factors like less educated parents, lower household income, and limited knowledge of vaccination schedules, especially in urban and peri-urban low-income areas. Barriers are primarily operational and informational rather than attitudinal rather than attitudes, which indicates that well-thought-out reminder systems, clear communication, and the proactive involvement of health workers can significantly improve in getting higher completion rates (Hobani & Alhalal, 2022; Khan et al., 2024; Ekezie et al., 2022).

Digital Health and SMS Reminder Interventions for Immunization

Digital immunization tools, including registries and SMS reminders, have demonstrated improvements in coverage and timeliness. These tools are particularly effective in resource-limited settings. Digital tracking systems in countries such as Bangladesh have enhanced the visibility of children who might otherwise be missed, supported faster identification of dropouts, and offered more accurate monitoring of coverage at local levels. Systematic reviews of SMS and call reminder interventions consistently reveal modest but meaningful increases in vaccination uptake compared to usual practice, and they are relatively inexpensive to implement. However, effectiveness is compromised by issues with phone connectivity and frequency of changes, language barriers, and the key need for active follow-up from health workers when automated reminders fail to produce action. This body of evidence supports the use of SMS reminders as part of a broader immunization support strategy, provided they are paired with strong engagement from community health workers and supplemented by supportive home-based follow-up. (Saha et al., 2023; Pavia et al., 2024; Jaleha et al., 2023; Wang et al., 2023).

Community Health Worker Centred Digital Support

The adoption of digital health tools and their proper usage depends on the designing process, which should be based not on the upper management but rather on the actual frontline workers' workflows, constraints, and needs. It has been found that among health care workers, if they perceive a technology to be useful, the efficiency of the tool will determine whether they will regularly log in and use it. The successfulness of digital interventions in low-resource settings is mainly characterized by their strikingly similar features: they are not increasing workload with loads of extra jobs, they can function in offline conditions or with very limited Internet access, they easily become part of the existing paper-based practices, and they are being supervised and supported by specialists who are always present in the community. The studies of vaccination registries in Africa and the Pacific have shown that systems are doomed when they generate more work, need constant troubleshooting, or are disconnected from the daily lives of health workers. This research thoroughly elaborates on the point that in order for a tool to be a support for community health workers, it must not only merely fit into their existing routines, but also provide them with clear and actionable information and make their hardest tasks—namely, identifying which kids are due for follow-up and knowing where to locate them—be less difficult, if not easy (Miguel Cruz et al., 2022; Saing et al., 2023; Wittenauer et al., 2023; Sheel et al., 2025).

Geospatial Technologies and Equity in Immunization

The spatial mapping and visualization of health data have emerged as crucial approaches not only to bring out the existing inequities but also to direct the resources to the neediest and most endangered communities. The immunization program staff is able to take advantage of the maps showing the areas with low coverage and thus facilitates efficient conduct of the outreach activities, thereby making sure that mobile clinics, health worker visits, and resources go where they are most needed. Dependable and simple visual tools like heat maps—which color code the displays according to the density of issues—have been effective in supporting decision-making and coordination whether they were done in immunization, perinatal health, or any other health domain. The review papers of the geographic information systems used in health care consistently point out that their use is very effective in disease monitoring and planning, while they also mention technical know-how, data quality, and infrastructure as the hurdles that need to be overcome in low-resource environments. For a rural health unit, even the simplest geospatial visualization—indicating at the barangay level where infants with missed or delayed vaccinations are concentrated—might allow managers and health workers to detect trends that would be masked by the statistics and thus make evidence-based decisions about the outreach capacity that they can afford to spread over the selected areas (Chaney & Mechael, 2025; Ngo-Bebe et al., 2025; van der Meer et al., 2022; Chandran & Roy, 2024).

Spatial Data Mining and Clustering Algorithms in Public Health Decision Support

The integration of advanced data mining techniques has revolutionized the field of health informatics by transforming raw geographic data into actionable intelligence. While traditional geospatial tools such as heatmaps provide a general visualization of data intensity, they often lack the mathematical precision required to guide specific field interventions. Recent literature emphasizes a shift toward Spatial Data Mining which utilizes algorithms to detect patterns and associations in health outcomes that are not random. Among these techniques, Density Based Spatial Clustering of Applications with Noise (DBSCAN) has emerged as a critical tool for identifying geographic pockets of vulnerability where under immunized infants are concentrated. Unlike partition based methods like K Means which require a predefined number of clusters, DBSCAN is capable of identifying clusters of arbitrary shapes and effectively filtering out isolated cases as noise. This capability is particularly vital in rural health settings where household distributions are uneven and traditional statistical averages may mask localized health inequities.

The application of clustering algorithms serves as the foundation for a Decision Support System by providing health administrators with prioritized outreach targets. Research indicates that the use of algorithmic centroids which represent the mathematical centers of identified clusters enables Rural Health Units to optimize the deployment of limited manpower and transportation resources. By mathematically ranking these clusters based on the density of overdue vaccination doses, the system moves beyond simple record keeping and into prescriptive analytics. This evidence based approach to outreach planning directly supports the Unified Theory of Acceptance and Use of Technology (UTAUT) by increasing the performance expectancy of health workers through reduced planning time and improved targeting accuracy. Ultimately, the literature suggests that the transition from descriptive visualization to density based clustering is a necessary evolution for community level health systems aiming to achieve herd immunity in underserved populations.

Immunization Information Systems, Data Use, and Privacy

The electronic immunization information systems are widely used in different parts of the world, yet studies indicate that their capabilities are rarely utilized due to the issues with design and implementation. Apart from that, many current systems are being defeated by poor interoperability between data sources, limited reminder functionality, incomplete life-course coverage, and data quality issues that negatively affect their usefulness for local decision-making. One such problem is that health workers frequently rely on paper records alongside or instead of electronic systems when the digital tools are slow, tedious, or disconnected from daily routines. When it comes to data privacy, public support for the sharing of health information is mainly dependent on the factors of trust, clear protection of personal data, transparency as to how the information will be used, and the opportunities for individuals to have some control over their own information; concerns are at their highest when it seems that data sharing is prioritizing commercial benefit over public health. For any new system that deals with sensitive information related to infants and families, adherence to data protection laws is always a must, and a real commitment to privacy has to be evident in the design of the system, governance, and communication with users (Vigezzi et al., 2025; Dombkowski et al., 2025; Baines et al., 2024).

Related Software

Synthesis (Citations)

The reviewed literature indicates that the persistent issue of incomplete infant vaccinations in developing regions is not primarily caused by a lack of vaccine supply, but rather by significant challenges in caregiver follow up and inefficient manual tracking systems. While digital interventions such as SMS reminders assist caregivers in remembering due dates and electronic records enable health personnel to monitor patient history, these instruments are most effective only when they are integrated with actual work patterns and customized for the constraints of rural health facilities. However, a critical gap exists as most current immunization systems are designed for national programs and lack the analytical capability to support local decision making in areas with poor connectivity and heavy workloads.

The literature further suggests that simple geospatial visualization is no longer sufficient for modern public health needs; instead, there is a shift toward spatial data mining to identify non random patterns in health data. By utilizing the DBSCAN algorithm, health systems can move beyond basic mapping and into prescriptive analytics, allowing for the mathematical identification of geographic pockets of vulnerability while filtering out isolated cases as noise. IMMUNICARE is designed to fill this research and operational void by serving as a practical, web based system that facilitates organized infant follow up, automated scheduling, and targeted outreach through density based spatial clustering. By providing actionable intelligence to midwives and barangay health workers, the system ensures that limited resources are directed toward high priority clusters while strictly observing community privacy and technical quality benchmarks based on the ISO IEC 25010 framework

Chapter 3

METHODOLOGY

Research Design

This study employs a quantitative developmental research design to develop and evaluate IMMUNICARE, a web-based Decision Support System (DSS) for infant immunization tracking and monitoring with spatial clustering for vaccination outreach. Design and Development Research (DDR) is a systematic methodology used for designing, developing, and evaluating technological systems intended to address identified real-world problems through iterative development and assessment processes (Aris et al., 2024). This framework guides the study's integration of software engineering development with a pre- and post-implementation evaluation structure, enabling the researchers to capture baseline conditions prior to deployment and to quantitatively measure system-driven changes in immunization data management following full implementation. This approach is appropriate for the study because it supports both software engineering development and quantitative evaluation of system effectiveness in a real healthcare setting.

The evaluative component of this study is organized into three parallel quantitative strands. The first strand assesses caregiver behavioral outcomes through a dual-data approach that combines objective vaccination attendance records with subjective caregiver self-report data collected through a Likert-scale survey instrument grounded in the Health Belief Model (HBM). This integration of behavioral and perceptual data is deliberate and methodologically necessary: attendance records alone reveal whether caregivers complied with vaccination schedules, but they cannot explain the cognitive and attitudinal factors that drove or hindered that behavior. By pairing objective compliance data with caregiver-reported cues to action, perceived benefits, and self-efficacy, the study achieves a richer and more explanatory understanding of how automated SMS reminders influence vaccination adherence — an analysis that neither data source could support independently.

The second strand measures the level of technological acceptance of IMMUNICARE among Rural Health Unit staff using a validated survey instrument grounded in the Unified Theory of Acceptance and Use of Technology (UTAUT), covering performance expectancy, effort expectancy, social influence, facilitating conditions, and behavioral intention to use. The third strand evaluates the technical quality of the system using the ISO/IEC 25010 software quality standard, assessing dimensions of functional suitability, usability, performance efficiency, reliability, and security, as appraised by IT experts with relevant backgrounds in software testing or health information systems.

Operationally, IMMUNICARE implements a barangay-scoped role-based workflow in which Barangay Health Workers (BHWs) create Draft infant registration records that undergo Midwife validation before becoming part of the official Master Infant Registry. Only validated records are processed for automated National Immunization Program (NIP) schedule generation, DBSCAN spatial clustering analysis, geospatial hotspot visualization, reporting, and SMS notification dispatch. The system also incorporates audit logging, OTP-authenticated caregiver portal access, and role-specific dashboards to support accountability, data integrity, and secure information management within participating Rural Health Units.

The study is designed to support immunization monitoring workflows across Rural Health Units within San Pedro City, Laguna, and covers infants from birth to twelve months of age, consistent with the complete first-year National Immunization Program schedule. However, due to time and operational constraints, pilot implementation and evaluation activities are limited to selected Rural Health Unit sites during the conduct of the study. The research does not employ a control group comparison or experimental randomization; rather, it relies on pre- and post-implementation measurements within the same implementation setting, supplemented by expert evaluation of system quality. This design reflects the operational and ethical constraints of conducting research within an active public health facility and is consistent with similar applied health informatics studies that prioritize system development and context-specific evaluation over controlled experimentation.

Sources of Data

The data utilized in this study are drawn from four primary sources, each corresponding to a distinct component of the research framework. The table below summarizes all data sources, their corresponding instruments, respondents or records, and the research questions they address.

Table 1

Source of data

(Pre-Development Assessment Questionnaire Respondents)

Description	Frequency	Percentage
Midwives	TBD	TBD
Nurses	TBD	TBD
Barangay Health Workers (BHWs)	TBD	TBD
Total Respondents	TBD	100%

The first source consists of data generated through a Pre-Development Assessment Questionnaire administered to RHU staff prior to the design and development of IMMUNICARE. This instrument is directed at midwives, nurses, and barangay health workers currently involved in immunization workflows at the pilot site. The questionnaire documents the baseline state of immunization management practices, including existing challenges in record-keeping, difficulties in identifying due and overdue infants, limitations of the present manual system, and the perceived needs of frontline health personnel. The data gathered from this instrument serve as a primary empirical basis for Research Question 1.

Table 2

Source of data

(Post-Implementation Survey Respondents (UTAUT and ISO/IEC 25010))

Description	Frequency	Percentage
Midwives	*TBD*	*TBD*
Nurses	*TBD*	*TBD*

Barangay Health Workers (BHWs)	*TBD*	*TBD*
System Administrators	*TBD*	*TBD*
IT Experts	*TBD*	*TBD*
Total Respondents	*TBD*	*100%*

The second source consists of primary data generated through the administration of structured survey instruments to two distinct respondent groups following system implementation. The first group consists of RHU staff directly involved in immunization workflows — specifically midwives, nurses, barangay health workers, and system administrators — who provide data on the level of technological acceptance of IMMUNICARE as measured through UTAUT constructs. The second group consists of IT experts who evaluate the technical quality of the system in accordance with the ISO/IEC 25010 standard. Both groups are purposively selected based on their direct relevance to the research objectives.

Table 3

Source of data

(HBM Caregiver Survey Respondents (Self-Report Data))

Description	Frequency	Percentage
Caregivers of enrolled infants — post-implementation	*TBD*	*TBD*
Total Caregiver Respondents	*TBD*	*100%*

The third source consists of primary data collected through the HBM-based Likert-scale caregiver survey (SOP 2), administered directly to caregivers of infants registered under the National Immunization Program at the pilot RHU following system implementation. This instrument captures subjective data on caregiver perceptions, attitudes, and self-reported compliance behaviors associated with the SMS reminder intervention. It operationalizes the HBM constructs of cues to action, perceived benefits, and caregiver compliance from the caregiver's own perspective. Because attendance records alone cannot account for the cognitive and attitudinal processes behind compliance behavior, this self-report data source is indispensable for fully answering Research Question 2.

Table 4

Source of data

(Vaccination Attendance Records (Objective Behavioral Data))

Description	Frequency	Percentage
Caregiver attendance records reviewed (pre-implementation)	*TBD*	*TBD*
Caregiver attendance records reviewed (post-implementation)	*TBD*	*TBD*
Total Records Extracted	*TBD*	*100%*

The fourth source consists of objective behavioral data extracted from official vaccination attendance records maintained at the pilot RHU. These records provide pre- and post-implementation vaccination attendance figures that serve as direct, observable measures of caregiver compliance behavior. Geographic and purok-level residential coordinate data of infants, together with caregiver contact information used for SMS notification dispatch, also constitute supplementary system inputs that inform the DBSCAN spatial clustering module and the automated scheduling engine of IMMUNICARE. Unlike the self-report caregiver survey, which captures perceptions, the attendance records capture what caregivers actually did — providing the behavioral anchor against which perceptual data from the survey are interpreted.

Research Instrument

Four distinct research instruments are employed in this study to gather empirical data across the developmental and evaluative phases of IMMUNICARE. All instruments are strictly grounded in the theoretical models, evaluation standards, and research questions established in Chapter 1, and are validated through expert review prior to administration. Specifically, the study utilizes: (1) a Pre-Development Assessment Questionnaire to document baseline workflows; (2) a UTAUT-based acceptance survey to measure user adoption; (3) an ISO/IEC 25010-based software quality evaluation checklist for technical appraisal; and (4) an HBM-based Likert-scale caregiver self-report survey to capture perceptual and attitudinal compliance data.

It is critical to distinguish between two categories of data collection instruments used in this study. The first category comprises survey instruments — structured questionnaires or checklists administered directly to respondents who provide their own input (i.e., the UTAUT survey, the ISO/IEC 25010 checklist, and the HBM caregiver self-report survey). The second category comprises a documentary record review — a structured data extraction form completed by the research team using existing RHU records rather than by respondents themselves. This distinction is methodologically significant because the two categories yield different types of data: survey instruments generate primary perceptual or evaluative data, while documentary record review yields objective behavioral data derived from official facility records.

Survey Instruments

The first instrument is a Pre-Development Assessment Questionnaire administered to RHU staff prior to the design and development of IMMUNICARE (Appendix A). This tool documents the baseline state of immunization management at the pilot site by gathering firsthand accounts of current workflows, operational challenges, and system limitations as experienced by frontline health personnel. The questionnaire covers the methods currently used to track due and overdue infants, the frequency and nature of defaulter identification difficulties, the time required to process vaccination records manually, and the perceived gaps in the existing system. Responses from this instrument directly address Research Question 1, specifically sub-questions 1.1 and 1.2, by providing practitioner-level evidence of baseline caseload handling and the operational context within which under-immunized clusters arise. The instrument also functions as a requirements-gathering tool that informs the system design phase, ensuring that IMMUNICARE is aligned with the documented needs of the health workers it is intended to support.

The second instrument is a UTAUT-based acceptance survey administered to RHU staff following system implementation (Appendix B). This instrument operationalizes the four core constructs of the Unified Theory of Acceptance and Use of Technology: performance expectancy, effort expectancy, social influence, and facilitating conditions, as well as the outcome variable of behavioral intention to use. Each construct is measured through a set of closed-ended items presented on a five-point Likert-type scale. Items are formulated to capture RHU staff perceptions of the system's usefulness, ease of use, peer and supervisory influence, and availability of supporting infrastructure within the context of their daily immunization workflows. This instrument directly addresses Research Question 3 and its sub-questions.

The third instrument is an HBM-based caregiver self-report survey administered to caregivers of enrolled infants following system implementation (Appendix D). This instrument operationalizes the Health Belief Model constructs of cues to action, perceived benefits, and self-reported compliance from the caregiver's own perspective. Items are presented on a five-point Likert-type scale and are designed to capture perceptual and attitudinal factors associated with SMS reminder–driven vaccination adherence. Because attendance records alone cannot explain the cognitive and motivational processes behind caregiver behavior, this self-report instrument is an indispensable complement to the objective record data, and together they fully address Research Question 2.

Documentary Record Review

The fourth instrument is a structured Documentary Record Review form used by the research team to extract objective behavioral data from official vaccination attendance records maintained at the pilot RHU (Appendix C). Unlike the survey instruments described above, this form is not administered to respondents; rather, it is completed by the research team through a systematic review of existing facility documentation. The form records pre- and post-implementation vaccination attendance figures, counts of infants classified as due, overdue, and defaulters, and geographic coordinate data used as inputs to the DBSCAN spatial clustering module. This instrument operationalizes the Health Belief Model construct of cues to action by tracking whether SMS reminders — as system-generated cues — are associated with measurable changes in caregiver attendance behavior. It directly addresses Research Question 2 and its sub-questions, providing the behavioral anchor against which self-reported perceptual data from the caregiver survey are interpreted.

Data Generation Procedure

The data generation procedure follows a sequential, phase-based process that mirrors the developmental and evaluative structure of the research design. The procedure unfolds across four distinct phases: baseline assessment, system development and validation, system deployment, and post-implementation data collection.

During the first phase, the research team conducts a comprehensive baseline assessment of the current manual immunization management workflow at the pilot Rural Health Unit. Two parallel activities characterize this phase. First, the Pre-Development Assessment Questionnaire (Appendix A) is administered to RHU staff — specifically midwives, nurses, and barangay health workers — to document their firsthand experience with existing immunization tracking procedures, the challenges they encounter in identifying due and overdue infants, and the limitations of the current manual system. Second, a structured documentary review of existing paper-based records is conducted to determine the caseload volume of infants classified as due, overdue, and recorded as defaulters, as well as the frequency of under-immunized clusters prior to system implementation. Together, these activities establish the pre-implementation baseline for Research Question 1 and provide the requirements data that inform the design of IMMUNICARE.

During the second phase, IMMUNICARE is developed in accordance with the functional and non-functional requirements identified during the baseline assessment and specified in the scope of the study, and is validated against the ISO/IEC 25010 benchmarks. Expert validation instruments are administered to content and technical experts to assess the validity of the system modules and survey instruments prior to full deployment. Revisions are made based on expert feedback to ensure that the system accurately reflects the immunization workflows of the pilot RHU and that the survey instruments possess adequate content validity.

During the third phase, IMMUNICARE is formally deployed at the pilot RHU. Authorized RHU staff, including midwives, nurses, and barangay health workers, are oriented on system navigation and role-based functions. The automated NIP schedule engine is activated, infant records are entered into the system, and geographic coordinate data are validated by authorized personnel. The SMS notification module begins dispatching automated vaccination reminders to registered caregivers, generating cues to action consistent with the Health Belief Model framework.

During the fourth phase, post-implementation data are collected. The UTAUT-based acceptance survey (Appendix B) is administered to RHU staff, and the ISO/IEC 25010-based quality evaluation checklist is administered to IT experts. The HBM caregiver self-report survey (Appendix D) is administered to caregivers of enrolled infants. Concurrently, the research team extracts post-implementation vaccination attendance data using the Documentary Record Review form (Appendix C) and compares these figures against baseline records. All data are encoded, organized, and subjected to appropriate statistical analysis to address each of the research questions.

Ethical Considerations

The conduct of this study adheres to established ethical principles governing research involving human participants and the management of sensitive health data. Given that IMMUNICARE processes personally identifiable information pertaining to infants and their caregivers, including residential coordinates, caregiver contact details, vaccination histories, and supporting documents such as birth certificates and prenatal records, the ethical management of such data is a foundational requirement of the study rather than a supplementary concern.

Informed consent is obtained from all human participants prior to their involvement in the study. RHU staff who participate in the Pre-Development Assessment Questionnaire and the UTAUT-based acceptance survey, as well as IT experts who complete the ISO/IEC 25010 quality evaluation checklist and caregivers who respond to the HBM self-report survey, are fully briefed on the purpose of the study, the voluntary nature of their participation, and their right to withdraw at any time without consequence. The Documentary Record Review is conducted exclusively from official RHU attendance records in accordance with the data governance policies of the pilot RHU and the City Health Office of San Pedro, Laguna; no direct research contact with caregivers is required for this activity.

Data privacy is enforced through the Role-Based Access Control mechanism embedded within IMMUNICARE, which restricts access to sensitive immunization information based on the assigned functional role of each user. Geospatial visualizations and spatial clustering outputs are restricted to aggregated barangay and purok levels, ensuring that no household-level GPS coordinates are exposed to unauthorized users. All data collected in this study are used exclusively for academic research and system evaluation purposes and are not shared with third parties for commercial or any non-research-related purposes. Findings are reported in aggregate or anonymized form to prevent the identification of individual participants or infant beneficiaries, and all data storage, handling, and reporting procedures comply with applicable data protection standards and the institutional policies of the San Pedro City Health Office.

Data Case Analysis

The data generated in this study are analyzed through a combination of descriptive statistics, survey scale analysis, and spatial cluster analytics, each aligned with the specific research questions and the theoretical frameworks established in Chapter 1.

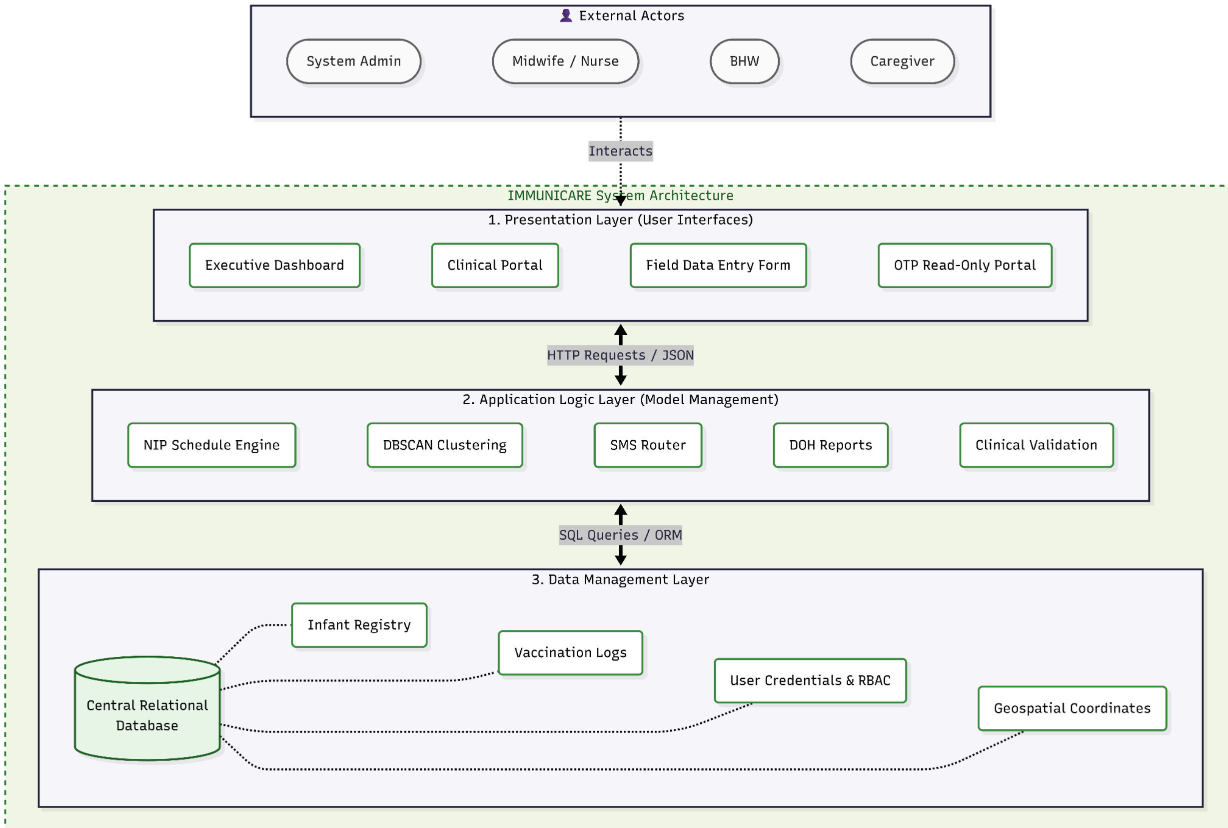
For Research Question 1, which addresses the baseline state of infant immunization management at the pilot RHU, data from two sources are analyzed in conjunction. Responses from the Pre-Development Assessment Questionnaire are analyzed descriptively to identify recurring themes in staff-reported workflow challenges, average manual processing times, and perceived system limitations. In parallel, frequency counts and percentage distributions are computed from pre-implementation immunization records for the total number of infants classified as due, overdue, and defaulters during the baseline period. The frequency of under-immunized geographic clusters identified through the DBSCAN algorithm prior to system optimization is also documented to establish the pre-implementation clustering profile of the RHU catchment area.

For Research Question 2, which examines the influence of automated SMS reminders on caregiver behavioral outcomes based on the Health Belief Model, analysis proceeds along two parallel tracks consistent with the dual-data approach. First, comparative analysis of vaccination attendance records — extracted through the Documentary Record Review — is conducted before and after system implementation; attendance rates and defaulter reduction figures are computed and compared to determine the extent to which the SMS reminder function is associated with measurable changes in caregiver compliance behavior. Second, weighted mean scores are computed for each HBM construct measured through the caregiver self-report survey — cues to action, perceived benefits, and caregiver compliance — to capture the attitudinal and perceptual factors that mediate observed behavioral outcomes. The integration of both analyses enables a comprehensive explanation of how automated reminders influence vaccination adherence.

For Research Question 3, which measures the level of technological acceptance of IMMUNICARE among RHU staff based on the UTAUT model, weighted mean scores are computed for each of the five constructs: performance expectancy, effort expectancy, social influence, facilitating conditions, and behavioral intention to use. Results are interpreted against a predefined scale to determine the overall level of acceptance, and construct-level scores are analyzed to identify which dimensions of system acceptance are most strongly endorsed.

For Research Question 4, which assesses the level of satisfaction and effectiveness of IMMUNICARE based on the ISO/IEC 25010 standard, weighted mean scores are computed for each of the five quality characteristics evaluated by IT experts: functional suitability, usability, performance efficiency, reliability, and security. Results are interpreted against the same scale used for the UTAUT analysis to determine the overall technical quality rating of the system and to identify specific quality dimensions that may require further improvement. All quantitative analyses are performed using appropriate statistical software, and results are presented in tabular form with accompanying narrative interpretation.

Proposed System Architecture



IMMUNICARE is architected as a web-based, multi-tier Decision Support System designed to operate within the infrastructural constraints of a Rural Health Unit in San Pedro City, Laguna. The system architecture is organized around the three core subsystems defined by Sprague and Carlson's (1982) DSS framework — data management, model management, and user interface — and is implemented as a three-layer technical stack comprising a presentation layer, an application logic layer, and a data management layer.

The presentation layer constitutes the user-facing component of the system and is accessible through standard web browsers, including Chrome, Firefox, and Edge, on any internet-connected device. This layer renders the role-based dashboards, data entry forms, geospatial heatmap visualizations, and report generation interfaces that correspond to the four user roles defined in the system: administrators, midwives or nurses, barangay health workers, and caregivers. The caregiver-facing interface is designed to provide secure, read-only access to infant vaccination records through a One-Time Password authentication mechanism, ensuring that sensitive health information is accessible only to verified caregivers without requiring the creation of permanent accounts.

The application logic layer constitutes the processing core of the system and houses four primary functional modules. The first is the automated NIP schedule engine, which computes vaccine due dates based on each infant's date of birth following the National Immunization Program schedule, enforces minimum dose interval rules, and applies catch-up logic when doses are missed. The second is the automated SMS notification module, which dispatches vaccination reminders and alerts to registered caregiver mobile numbers based on computed due dates, functioning as system-generated cues to action consistent with the Health Belief Model framework. The third is the DBSCAN spatial clustering module, which processes the residential coordinate data of under-immunized infants to identify geographic clusters of vaccination gaps, computes geographical centroids for each cluster to guide outreach planning, and generates geospatial heatmap visualizations at the barangay and purok levels. The fourth is the Department of Health report generation module, which produces standardized Monthly Form 1 and Annual Form A1 immunization summaries for submission to higher health offices.

The data management layer consists of a server-hosted relational database management system that stores and manages all immunization records, geographic coordinate data, caregiver contact details, and system-generated outputs. The database architecture is designed to meet the security and reliability benchmarks of ISO/IEC 25010, with Role-Based Access Control enforced at the database level to ensure that each user type can access only the data and functions appropriate to their assigned role. Geospatial data are stored and processed at the aggregated barangay and purok levels within the database, with no household-level GPS coordinates exposed through any user-facing interface. The system requires a stable internet connection for real-time data entry, SMS notification dispatch, and spatial clustering computation; no offline functionality or native mobile application is included within the current scope of the system.